



COS30017

Software Development for Mobile Devices

Learning Summary Report

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Self-Assessment Details

The following checklists provide an overview of my self-assessment for this unit.

Self-Assessment Statement

	Pass (D)	Credit (C)	Distinction (B)	High Distinction (A)
Self-Assessment		✓		

Declaration

I declare that this portfolio is my individual work. I have not copied from any other student's work or from any other source except where due acknowledgment is made explicitly in the text, nor has any part of this submission been written for me by another person. Failure to meet this requirement will result in a failing grade for the unit.

Failure to provide the source code for any task submission will result in that task not being assessed, even if the task is included in PDF format.

Signature: **Nguyễn Chí Cường**

Portfolio Overview

This portfolio includes work that demonstrates that I have achieved all Unit Learning Outcomes for COS30017 Software Development for Mobile Devices to a **Credit** level.

Throughout the unit, I have consistently applied core mobile development concepts and demonstrated a solid understanding of Android application design, development, and implementation.

Across multiple assignments, I developed functional Android applications. These tasks required me to implement user interfaces using XML layouts, manage activity and fragment lifecycles, handle user input, and integrate data persistence using databases. I also applied debugging techniques to resolve runtime issues and improve application performance.

My work reflects a good level of competency in designing structured, user-friendly mobile applications while following recommended development practices. I was able to connect theoretical knowledge with practical implementation, particularly in areas such as UI design, data handling, and application architecture.

Overall, the tasks included in this portfolio demonstrate my ability to meet all learning outcomes to a good standard, justifying my application for a **Credit grade**.

Task Summary

To demonstrate my learning in this unit, I would like the following tasks to be considered part of my portfolio:

- Assignment 1 – Gymnastics App (**Completed**)
- Assignment 2 – Car Rental App (**Completed**)
- Assignment 3 – Novel Library App (**Completed**)

Reflection

The most important things I learnt:

The most important things I learned in this unit include how to design and develop Android applications. I gained a strong understanding of activity and fragment lifecycles, UI design using XML, and how to handle user interactions effectively. I also learned how to store data locally using databases and how to debug common issues in Android applications.

The things that helped me most were:

The things that helped me the most were:

- Online documents and tutorial: this allows me to look deeper into how things work and what I can use for my app.
- Weekly lab exercises: This allows me to get a strong grasp on the basics of Android development and the basic functions.

I found the following topics particularly challenging:

I found that setting up fragments at first to be particularly challenging, as managing their lifecycle and how they interact with activities correctly was quite confusing.

I found the following topics particularly interesting:

I found creating and designing layout to improve user experience particularly interesting and rewarding. Creating a good UI satisfy me very much.

I feel I learnt these topics, concepts, and/or tools really well:

I felt like I have learned how to manage fragments and UI design particularly well because it has my interest since the start of the semester.

I still need to work on the following areas:

I still need to learn how to optimize my app more with less lag between transitions and other optimizations problem.

My progress in this unit was ...:

My progress in this unit has been consistent. I completed all required tasks and engaged with the learning materials regularly. While I faced challenges in some topics, I was able to overcome them through practice and independent learning. This consistent effort helped me build a better understanding of mobile development concepts.

This unit will help me in the future:

This unit helps improve essential skills for my future studies and career in software development. Mobile application development is highly relevant in the industry, and the knowledge gained here—such as UI design, data handling, and debugging—will be directly applicable in real-world projects.

If I did this unit again I would do the following things differently:

- Start assignments earlier to allow more time for implementation and testing
- Spend more time practicing advanced concepts like architecture patterns
- Ask more questions during labs to clarify difficult topics sooner
- Explore additional features beyond requirements to improve my skills